Cambridge Assessment International Education

# Cambridge O Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*2538130325*

**ADDITIONAL MATHEMATICS** **4037/21**

Paper 2 **May/June 2020**

**2 hours**

You must answer on the question paper.

No additional materials are needed.

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

**INFORMATION**

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages. Blank pages are indicated.


© UCLES 2020

**[Turn over**

## *Mathematical Formulae*

### 1. ALGEBRA

*Quadratic Equation*

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

*Binomial Theorem*

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \ldots + \binom{n}{r}a^{n-r}b^r + \ldots + b^n$$

where $n$ is a positive integer and $\binom{n}{r} = \frac{n!}{(n-r)!r!}$

*Arithmetic series*

$$u_n = a + (n-1)d$$

$$S_n = \frac{1}{2}n(a+l) = \frac{1}{2}n\{2a + (n-1)d\}$$

*Geometric series*

$$u_n = ar^{n-1}$$

$$S_n = \frac{a(1-r^n)}{1-r} \quad (r \neq 1)$$

$$S_\infty = \frac{a}{1-r} \quad (|r| < 1)$$

### 2. TRIGONOMETRY

*Identities*

$$\sin^2 A + \cos^2 A = 1$$
$$\sec^2 A = 1 + \tan^2 A$$
$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

*Formulae for $\Delta ABC$*

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$\Delta = \frac{1}{2}bc\sin A$$

© UCLES 2020

**1** Variables $x$ and $y$ are such that, when $\sqrt[4]{y}$ is plotted against $\frac{1}{x}$, a straight line graph passing through the points (0.5, 9) and (3, 34) is obtained. Find $y$ as a function of $x$. [4]

**2** **(a)** Write $9x^2 - 12x + 5$ in the form $p(x-q)^2 + r$, where $p$, $q$ and $r$ are constants. [3]

**(b)** Hence write down the coordinates of the minimum point of the curve $y = 9x^2 - 12x + 5$. [1]

**3 DO NOT USE A CALCULATOR IN THIS QUESTION.**

$$\mathrm{p}(x) = 15x^3 + 22x^2 - 15x + 2$$

**(a)** Find the remainder when $\mathrm{p}(x)$ is divided by $x + 1$. [2]

**(b) (i)** Show that $x + 2$ is a factor of $\mathrm{p}(x)$. [1]

**(ii)** Write $\mathrm{p}(x)$ as a product of linear factors. [3]

© UCLES 2020

**4** **(a)** In an examination, candidates must select 2 questions from the 5 questions in section A and select 4 questions from the 8 questions in section B. Find the number of ways in which this can be done. [2]

**(b)** The digits of the number 6 378 129 are to be arranged so that the resulting 7-digit number is even. Find the number of ways in which this can be done. [2]

**5** The vectors $\mathbf{a}$ and $\mathbf{b}$ are such that $\mathbf{a} = \alpha\mathbf{i} + \mathbf{j}$ and $\mathbf{b} = 12\mathbf{i} + \beta\mathbf{j}$.

**(a)** Find the value of each of the constants $\alpha$ and $\beta$ such that $4\mathbf{a} - \mathbf{b} = (\alpha + 3)\mathbf{i} - 2\mathbf{j}$. [3]

**(b)** Hence find the unit vector in the direction of $\mathbf{b} - 4\mathbf{a}$. [2]

**6** Find the values of $k$ for which the line $y = kx - 7$ and the curve $y = 3x^2 + 8x + 5$ do not intersect. [6]

© UCLES 2020

**7** **(a)** Solve the simultaneous equations

$$10^{x+2y} = 5,$$
$$10^{3x+4y} = 50,$$

giving $x$ and $y$ in exact simplified form. [4]

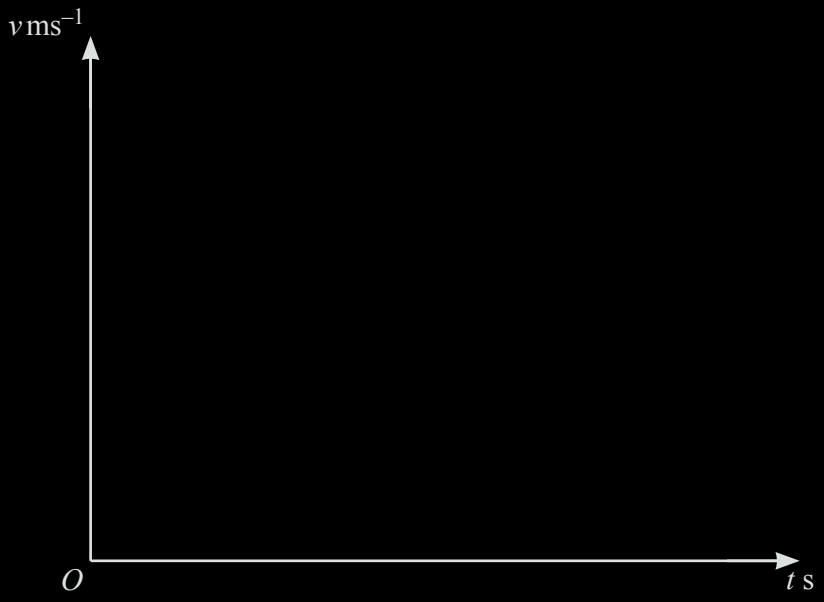
**(b)** Solve $2x^{\frac{2}{3}} - x^{\frac{1}{3}} - 10 = 0$. [3]

**8** **(a)** Expand $(2-x)^5$, simplifying each coefficient. [3]

**(b)** Hence solve $\dfrac{e^{(2-x)^5} \times e^{80x}}{e^{10x^4+32}} = e^{-x^5}$. [4]

**9** A particle travels in a straight line. As it passes through a fixed point $O$, the particle is travelling at a velocity of $3\,\text{ms}^{-1}$. The particle continues at this velocity for 60 seconds then decelerates at a constant rate for 15 seconds to a velocity of $1.6\,\text{ms}^{-1}$. The particle then decelerates again at a constant rate for 5 seconds to reach point $A$, where it stops.

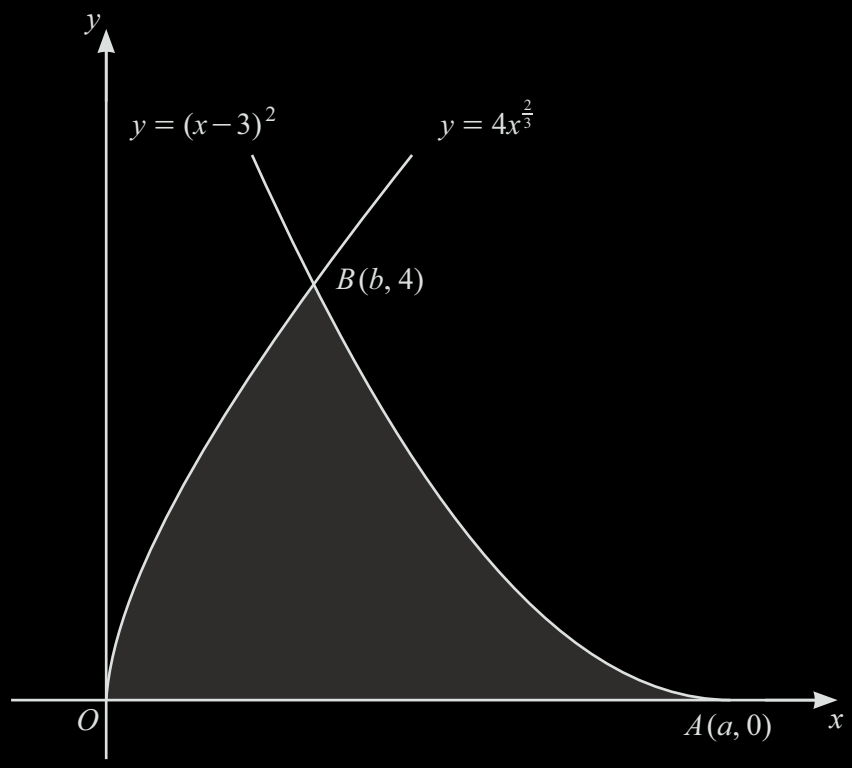
**(a)** Sketch the velocity-time graph for this journey on the axes below. [3]




**(b)** Find the distance between $O$ and $A$. [3]

**(c)** Find the deceleration in the last 5 seconds. [1]

**10**



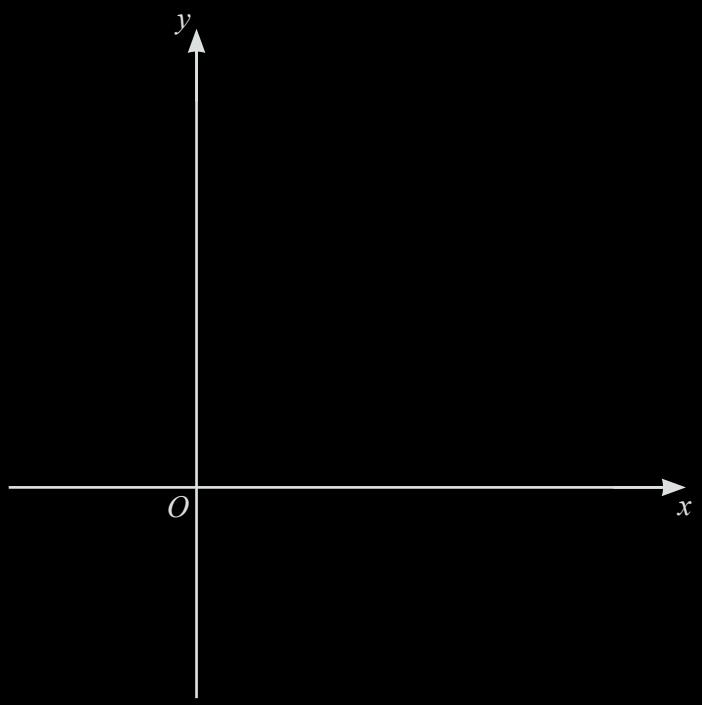

The diagram shows part of the graphs of $y = 4x^{\frac{2}{3}}$ and $y = (x-3)^2$. The graph of $y = (x-3)^2$ meets the $x$-axis at the point $A(a, 0)$ and the two graphs intersect at the point $B(b, 4)$.

**(a)** Find the value of $a$ and of $b$. [2]

**(b)** Find the area of the shaded region. [5]

**11** The function f is defined by $\mathrm{f}(x) = \ln(2x+1)$ for $x \geqslant 0$.

**(a)** Sketch the graph of $y = \mathrm{f}(x)$ and hence sketch the graph of $y = \mathrm{f}^{-1}(x)$ on the axes below. [3]




The function g is defined by $\mathrm{g}(x) = (x-4)^2 + 1$ for $x \leqslant 4$.

**(b)** **(i)** Find an expression for $\mathrm{g}^{-1}(x)$ and state its domain and range. [4]

**(ii)** Find and simplify an expression for $\mathrm{fg}(x)$. [2]

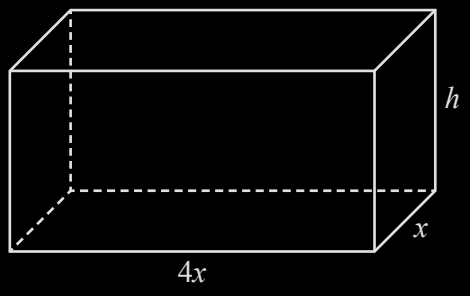
**(iii)** Explain why the function gf does not exist. [1]

**12** **(a)** Find the $x$-coordinates of the stationary points of the curve $y = e^{3x}(2x+3)^6$. [6]

**(b)** A curve has equation $y = f(x)$ and has exactly two stationary points. Given that $f''(x) = 4x - 7$, $f'(0.5) = 0$ and $f'(3) = 0$, use the second derivative test to determine the nature of each of the stationary points of this curve. [2]

© UCLES 2020

**(c)** In this question all lengths are in centimetres.




The diagram shows a solid cuboid with height $h$ and a rectangular base measuring $4x$ by $x$. The volume of the cuboid is $40\,\text{cm}^3$. Given that $x$ and $h$ can vary and that the surface area of the cuboid has a minimum value, find this value. [5]

**BLANK PAGE**

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of the Cambridge Assessment Group. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which itself is a department of the University of Cambridge.

© UCLES 2020